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MIL-D-70826A (AR)
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SUPERSEDING
MIL-D-70826 (AR)
09 September 1988

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, PA537 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, packaging, fabrication, manufacture and loading of one type of electric detonator designated as Detonator, Electric, PA537.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1375

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SPECIFICATIONS

FEDERAL

A-A-208 - Ink, Marking Stencil, Opaque (Porous and Non-Porous Surfaces).

MILITARY

MIL-P-116 - Preservation, Methods Of
MIL-B-117 - Bag, Sleeve and Tubing, Interior Packaging
MIL-G-45204 - Gold Plating, Electrodeposited
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-202 - Electronic and Electrical Component Parts, Test Methods for
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Test For
MIL-STD-650 - Explosive Sampling, Inspection and Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
12550926 - Detonator, Electric PA537
9326705 - Microdetonator Clip Assembly
12550826 - Test Circuit

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12550757 - Ballistic Environment Test Requirement of M762/M767 Components and Subassemblies

(Copies of other Government documents, drawings, and publications, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)".)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12550926, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Moisture content.

3.3.1 Lead azide. The moisture content of the lead azide at the loading station at the time of loading the detonators shall not exceed 0.30 percent, when tested as specified in 4.5.2.1.

3.3.2 HMX. The moisture content of the HMX at the time and place of loading the detonators shall not exceed 0.30 percent, when tested as specified in 4.5.2.2.

3.4 Output. The detonator shall not dud, shall fire on the first attempt, and shall make a minimum depression of 0.005 inches

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in the steel witness plate when tested as specified in 4.5.3.1.

3.5 Waterproofness. The detonators shall be waterproof as evidenced by normal functioning after being immersed in water maintained at 70 degrees Fahrenheit (F) plus or minus 10 degrees F, at a depth of 2 inches minimum, for not less than 48 hours.

3.6 Environmental

3.6.1 Resistance between electrode and cup, thermal shock and vibration. The detonators shall comply with the resistance requirement of Drawing 12550926 and with output requirements of 3.4 after being exposed to thermal shock and vibrations as specified in 4.5.4.1.

3.6.2 Cold temperature. The detonator assembly shall comply with output requirements of 3.4 when conditioned at -65 ± 5 degrees F for two hours and tested as specified in 4.5.4.2.

3.6.3 Hot temperature. The detonator assembly shall comply with output requirements of 3.4 when conditioned at +165 ± 5 degrees F for two hours and tested as specified in 4.5.4.3.

3.6.4 Shock G. The detonator assembly shall comply with the requirements of 4.5.8 after being subjected to a 30,000 g setback force an specified in 4.5.4.4. The detonator assembly shall be conditioned at 70 degrees ± 10 degrees F for two hours minimum prior to testing.

3.7 Shorting during shipping. The detonator shall be shorted when assembled into the microdetonator clip assembly (Dwg. 9326705) as evidence by a resistance of 0.5 ohm maximum between the detonator electrode and the cup.

3.8 First article. When specified in the contract or purchases order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.9 Change of qualified product. After first article inspection, the manufacturer shall notify the purchaser prior to the implementation of any change which may affect performance, quality, reliability or interchangeability. Such notification shall include a thorough description of the proposed change and a suggested test plan designed to demonstrate that the change will not adversely affect performance, quality, reliability or interchangeability and that the changed product will continue to meet the specification requirements. The purchaser will review the

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proposed change and test plan and advise the manufacturer of the testing required to obtain approval of the proposed change. Upon completion of the prescribed tests, the result shall be submitted to the purchaser for review. Based upon the review the purchaser will approve or disapprove the change and notify the manufacturer. At the manufacturer's option, devices incorporating the change may be manufactured and tested prior to approval, however, all shipments of these changed devices shall be withheld until approval is granted by the purchaser. Changes representative of those which are subject to this requirement or any design change in the manufacturing processing such as:

a. Package structure and material changes or changes to the geometry.

b. Changes to process flow charts, assembly flow chart and testing flow charts.

c. Conductor materials.

d. Assembly operations move from one line or building to another.

3.10 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. Cleaning methods used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.3.2 Inspection to be performed. See MIL-A-48078 and TABLE I herein.

4.3.3 Rejection. See MIL-A-48078 and TABLE I herein.

4.3.4 Component qualification. Qualification of the component supplied to this specification is not complete until it has been submitted to and passed additional tests performed by the purchaser. These tests may include 30,000 G acceleration tests, 30,000 RPM spin tests and aging tests to evaluate long term storage characteristics. Coordination of testing and test reporting shall be the responsibility of the purchaser.

4.3.4.1 Re-qualification. Re-qualification of changed component(s) may depend on the result circuit and application analysis conducted as a result of the change(s).

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of electric detonators shall contain:

a. Component parts from one lot interfix number from one manufacturer.

b. HMX from one lot number only.

c. Lead azide of the same type from not more than two lots from the same manufacturer, in accordance with the same specification and specification revision and which have subsequently been subjected to uniformly controlled drying and processing conditions by the manufacturer. If more than one lead azide lot is used the lots shall not differ by more than three serial lot numbers.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with TABLE II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-D-70826A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup	1 OF 1	9245697
			NEXT HIGHER ASSEMBLY: 9257046

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness through bottom	Level IV	3.2	Gage
102	Outside diameter	Level IV	3.2	Gage
103	Concentricity of outside diameter with inside diameter	Level IV	3.2	Gage
104	Dented or distorted	Level IV	3.2	Visual
Minor				
201	Inside diameter, minimum	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in quantities as indicated.

Part Description	Drawings	Quantity
Cup	9245697	50
Plug Assembly	9245694	50
Bridge Plug and Ignition Assembly	12550787	50
Charged Cup	9257046	50
Detonator Assembly	12550926	380

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TALBE I . First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
	Detonator assembly and components	SHEET 1 OF 1	See Below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH / INSPECTION METHOD REFERENCE
	Cup (Dwg. 9245697)		
	Examination for defects	50	3.2 / 4.4.2.1
	Plug Assembly (Dwg. 9245694)		
	Examination for defects	50	3.2 / 4.4.2.3
	Pull test of electrode	50	3.2 / 4.5.5
	Insulation resistance between electrode and header	50	3.2 / 4.5.6
	Bridge Plug and Ignition Assembly (Dwg. 12550787)		
	Examination for defects	50	3.2 / 4.4.2.4
	Detonator Assembly (Dwg. 12550926)		
	Examination for defects	Note 1	3.2 / 4.4.2.6
	Waterproofness	80	3.5 / 4.5.3.2
	Functioning output, dent test	200	3.4 / 4.5.3.1
	Environmental Tests		
	a) Resistance between electrode and cup, thermal shock and vibration	25-0-1	3.6.1 / 4.5.4.1
	b) Cold temperature conditioning	25-0-1	3.6.2 / 4.5.4.2
	c) Hot temperature conditioning	25-0-1	3.6.3 / 4.5.4.3
	d) Shock, G test	25-0-1	3.6.4 / 4.5.4.4

NOTES:

1. The entire quantity of detonators required for the tests below shall be subjected to the examination for defects prior to Functional testing.

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TABLE II Attributes sampling inspection

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more failures/defects for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.6). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-D-70826A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Bridgewire	1 OF 1	12550789
			NEXT HIGHER ASSEMBLY: 12550788

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Platinum wire diameter	See Note	3.2	Gage
Minor 201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

1. Five (5) measurements shall be made in the first two (2) feet of wire of each spool, and at least once per shift thereafter.

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CLASSIFICATION OF CHARACTERISTICS

MIL-D-70826A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Charged cup	1 OF 1	9257046
			NEXT HIGHER ASSEMBLY: 12550926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Height of total charge	Level IV	3.2	Gage
102	Moisture content of lead azide	4.4.3.1	3.3.1	4.5.2.1
103	Moisture content of HMX	4.4.3.1	3.3.2	4.5.2.2
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-D-70826A (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.3	Plug Assembly	SHEET 1 OF 1	9245694
			NEXT HIGHER ASSEMBLY: 12550788

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pull test of electrode	50-0-1	3.2	4.5.5
102	Insulation resistance test	100%	3.2	4.5.6
103	Outside diameter, max	Level IV	3.2	Gage
104	Bead cracked	Level IV	3.2	Visual
105	Gold plating	Note 1	3.2	4.5.9
Minor				
201	Length of electrode below surface of header	Level V	3.2	Gage
202	Thickness of header	Level V	3.2	Gage
203	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

1. Sampling and rejection shall be in accordance with MIL-G-45204.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Microdetonator Clip Assembly, (After Assembly of Detonator)	1 OF 1	9326705
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
special a	Detonator not shorted	100%	3.7	Current limited ohmmeter (1 milliamp, max.)
Major 101	Detonator electrode protruding above flush of clip assembly	100%	3.7	Visual
102	Detonator cup protruding above flush of clip assembly	100%	3.7	Visual
103	Detonator inverted in clip	100%	3.7	Visual
Minor 201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables and in 4.4.3.1.

4.4.3.1 Moisture content.

Material	Requirement	Test Paragraph
Lead azide	3.3.1	4.5.2.1
HMX	3.3.2	4.5.2.2

The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one (1) sample of each material from each eight (8) hours of detonator production. Composite samples shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, that quantity of material represented by the sample shall be rejected. If detonators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample, together with all detonators loaded with the non-conforming material, shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the First Article table or Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. See Section 6 of the MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply.

4.5.2 Moisture content.

4.5.2.1 Lead azide. The moisture content of the lead azide shall be determined in accordance with method 101.5 of MIL-STD-650 except that a two (2) gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

4.5.2.2 HMX. The moisture content of HMX shall be determined as specified in 4.5.2.1.

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Bridge Plug and Ignition Assembly	1 OF 1	12550787
			NEXT HIGHER ASSEMBLY: 12550926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Spot charge missing	Level IV	3.2	Visual
102	Cracks or voids in spot change	Level IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Detonator Assembly	1 OF 1	12550926
			NEXT HIGHER ASSEMBLY: Final

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Cup cracked or split	100%	3.2	Visual
b	Explosive on exterior of cup	100%	3.2	Visual
Major				
101	Length of electrode	Level IV	3.2	Gage
102	Length, crimped end to base	Level IV	3.2	Gage
103	Depth to spherical radius	Level VI	3.2	Gage
104	Large diameter	Level IV	3.2	Gage
105	Spherical radius missing, deformed or scratched	Note 2	3.2	Visual
106	Anvil spherical radius verification	Level IV	3.2	Gage
107	Sealant not present 360 degrees	Level IV	3.2	Visual
108	Decrease of sealant on electrode beyond sealed area	Level IV	3.2	Visual
109	Waterproofness test	50-0-1 Note 1	3.5	4.5.3.2
110	Functioning output (dent test)	200-1-2 Note1	3.4	4.5.3.1
111	Environmental test			
	a)Resistance between electrode and cup	100%	3.2	4.5.4.1a
	b)Cold temperature conditioning up	25-0-1	3.2	4.5.4.2
	c)Hot temperature conditioning	25-0-1	3.2	4.5.4.3
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES: 1. Any detonator that fails to function on first attempt or fails to produce the required output shall be classed defective.
2. Tooling dimensions are verified prior to manufacturing each lot and at the end of each lot.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Sealed Bag	1 OF 1	See paragraph 5.1
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	None defined			
Major				
101	Heat seal test	Note 1	3.2	4.5.10
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:
1. Sampling and rejection shall be in accordance with MIL-P-116.

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4.5.3 Functioning output tests.

4.5.3.1 Functioning (steel plate). The detonator assemblies shall be temperature conditioned at 70 ± 10 degrees Fahrenheit (F) for two (2) hours minimum (unless otherwise specified) prior to performing the functioning tests. The tests shall be performed as specified in MIL-STD-331, Test 301.1 except that the witness plate shall be 1/8 inch minimum and may optionally be round, 1.22 - .01 dia. Witness plate hardness shall be Rockwell B70 to B95. The base of the detonator shall be placed directly against the center of the witness plate and electrical contact made through a clip which will provide a minimum confinement of the detonator. One lead shall be affixed to the detonator cup and the other to the electrode. The detonator shall be fired by applying energy from a test circuit made to Drawing 12550826. The depression in the witness block shall be measured using a dial indicator equipped with a probe having a 0.003 ± 0.001 inch spherical radius and in accordance with the requirements of 3.4

4.5.3.2 Waterproofness output. The detonator assemblies shall be immersed in water at a two (2) to three (3) inch depth for 48 hours minimum. The water temperature shall be maintained at 70 ± 10 degrees F from the immersion period. At the end of the immersion time, the detonator assemblies shall be removed from the water, assembled onto the test fixture and fired by applying energy from a test circuit made of Drawing 12550826. The depression in the witness block shall be measured using a dial indicator equipped with a probe having a 0.003 ± 0.001 inch spherical radius and in accordance with the requirements of 3.4.

The test shall be performed as specified in MIL-STD-331, Test 301.1 except that the witness plate shall be 1/8 inch minimum and may optionally be round, 1.22 - .01 dia. Witness plate hardness shall be Rockwell B70 to B95. The base of the detonator shall be placed directly against the center of the witness plate and electrical contact made through a clip which will provide a minimum confinement of the detonator. One lead shall be affixed to the detonator cup and the other to the electrode.

4.5.4 Environmental tests.

4.5.4.1 Resistance between electrode and cup, thermal shock and vibration. The detonator assemblies shall be tested in the following sequence.

a. Measure resistance between the electrode and cup in accordance with 4.5.8.

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b. Condition for two (2) hours minimum as per 3.6.2.

c. Remove from cold chamber and immediately transfer to the hot chamber.

d. Condition for two (2) hours minimum as per 3.6.3.

e. Remove from hot chamber and immediately transfer to the vibrator.

f. Vibrate for 15 minutes minimum in a sweep covering frequency range of 25 - 500 - 25 cycles per second parallel to the transverse orthogonal axis with base charge down and a 2 ± 0.2 g's peak.

g. Measure resistance between electrode and cup in accordance with 4.5.8.

h. The detonators shall be functioned as specified in 4.5.3.1.

4.5.4.2 Cold temperature. The detonator assemblies shall be conditioned to -65 ± 5 degrees F for two (2) hours minimum. The detonators, maintained at the conditioning temperature, shall be functioned as specified in 4.5.3.1.

4.5.4.3 Hot temperature. The detonator assemblies shall be conditioned to +165 ± 5 degrees F for two (2) hours minimum. The detonator assemblies, maintained at the conditioning temperature, shall be functioned as specified in 4.5.3.1.

4.5.4.4 Shock, G. The detonator assemblies, maintained at the conditioning temperature, shall be subjected to a 30,000g setback force in accordance with the ballistic environment test requirements of M762/M767 components and subassemblies, 12550757. After completion of the test, the detonator assemblies shall be tested as specified in 4.5.8. Test samples shall not be returned to the lot.

4.5.5 Pull test of electrode. The plug assembly shall be assembled to the equipment and tests in accordance with Method 211 of MIL-STD-202. The electrode and header shall be observed for compliance with the requirement. This is a destructive test. Test samples shall not be returned to the lot.

4.5.6 Insulation resistance between the electrode and header. The plug assembly shall be connected to the equipment and the required voltage applied for not less than one (1) nor more than

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Each intermediate carton, set-up box or comparable container, after blunting corners, shall be placed in a water and vapor proof bag manufactured in accordance with Type I, Class E of MIL-B-117. The bag shall be closed. All excess air shall be exhausted from within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two (2) subsequent seals, if required. The seal shall comply with the heat seal requirements of MIL-P-116.

5.1.2 Level C. Unless otherwise specified, the detonators shall be packaged in accordance with 5.1.1, except that the barrier bag is not required.

5.2 Packing.

5.2.1 Level B. A suitable number of detonators packaged as specified in 5.1.1, shall be packed in an outer box in accordance with Code of Federal Regulations Title 49: Transportation of Explosives and Other Dangerous Articles, Paragraph 173.66.

5.2.2 Level C. Unless otherwise specified, detonators shall be packed as specified in 5.2.1.

5.3 Marking. Each inner container and intermediate carton set-up box or comparable container shall be printed, stamped or labeled and each barrier bag printed or stamped in accordance with the following information and instructions:

a. (Number) Detonator, Electric, PA 537.

   Lot No. Date Packed(Mo/Yr)

b. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4 inch high applied with black waterproof ink. Stencil ink shall conform to A-A-208.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with Drawing 8796552. The DOT designation shall be: Electric Blasting Caps - Handle Carefully (see NOTE).

e. NOTE: "This number of caps packed in one (1) outer pack shall be shown in the marking."

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6.6 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.7 Subject term (key word) listing.

Moisture content
Resistance
Shock, G
Waterproofness

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-0006)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-70826A (AR)	940602

3. DOCUMENT TITLE DETONATOR, ELECTRIC, PA537 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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two (2) seconds. The equipment shall be observed to determine compliance with the requirements. Any plug assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.5.7 Resistance between electrode and header. The bridge assembly shall be connected to the equipment and the bridge resistance shall be measured when a current limited to ten (10) milliampere is applied. Any bridge assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.5.8 Resistance between electrode and cup. The detonator assembly shall be connected to the equipment and the resistance shall be measured between the electrode and the cup when a current limited to ten (10) milliampere is applied. Any detonator assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.5.9 Gold plating of pin in plug assembly (See Dwg 9245694 covering detail to Dwg 12550926). Sampling, acceptance criteria and test procedure for this test shall be as specified in MIL-G-45204.

4.5.10 Heat seal test of sealed bag.
Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-116.

4.5.11 Tool control for spherical radius. The contractor shall verify at the beginning and end of each detonator lot produced that the anvil used meets the spherical radius requirements of (Dwg. 12550926). Non-compliance with the above requirements shall result in the rejection of the lot.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level B. Detonators shall be packaged in accordance with Code of Federal Regulations Title 49, Para 100-17. In addition, each detonator shall be electrically shorted during shipment and storage by means of a metallic connection from the detonator pin to the cup by assembly into microdetonator clip assembly, Drawing 9326705. When destined for shipment and uncovered storage in excess of 180 days from the loading date shown on the packing box to the date of the assembly into the end item, the following shall be included:

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5.4 Shipping. When detonators from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The detonator assembly covered by the specification is intended for use in munition assemblies.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1)

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Distribution of ammunition data cards, inspection and test results, process controls. Distribution of ammunition data cards, inspection and test results, process controls shall be sent to:

Commander
ARDEC
ATTN: SMCAR-QAR-I
Picatinny Arsenal, New Jersey 07806-5000.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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TALBE I . First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Detonator assembly and components	1 OF 1	See Below

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Cup (Dwg. 9245697)			
	Examination for defects	50	3.2	4.4.2.1
	Plug Assembly (Dwg. 9245694)			
	Examination for defects	50	3.2	4.4.2.3
	Pull test of electrode	50	3.2	4.5.5
	Insulation resistance between electrode and header	50	3.2	4.5.6
	Bridge Plug and Ignition Assembly (Dwg. 12550787)			
	Examination for defects	50	3.2	4.4.2.4
	Detonator Assembly (Dwg. 12550926)			
	Examination for defects	Note 1	3.2	4.4.2.6
	Waterproofness	80	3.5	4.5.3.2
	Functioning output, dent test	200	3.4	4.5.3.1
	Environmental Tests			
	a) Resistance between electrode and cup, thermal shock and vibration	25-0-1	3.6.1	4.5.4.1
	b) Cold temperature conditioning	25-0-1	3.6.2	4.5.4.2
	c) Hot temperature conditioning	25-0-1	3.6.3	4.5.4.3
	d) Shock, G test	25-0-1	3.6.4	4.5.4.4

NOTES:

1. The entire quantity of detonators required for the tests below shall be subjected to the examination for defects prior to Functional testing.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup	1 OF 1	9245697
			NEXT HIGHER ASSEMBLY: 9257046

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness through bottom	Level IV	3.2	Gage
102	Outside diameter	Level IV	3.2	Gage
103	Concentricity of outside diameter with inside diameter	Level IV	3.2	Gage
104	Dented or distorted	Level IV	3.2	Visual
Minor				
201	Inside diameter, minimum	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-D-70826A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Bridgewire	1 OF 1	12550789
			NEXT HIGHER ASSEMBLY: 12550788

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Platinum wire diameter	See Note	3.2	Gage
Minor 201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

1. Five (5) measurements shall be made in the first two (2) feet of wire of each spool, and at least once per shift thereafter.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Plug Assembly	1 OF 1	9245694
			NEXT HIGHER ASSEMBLY: 12550788

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pull test of electrode	50-0-1	3.2	4.5.5
102	Insulation resistance test	100%	3.2	4.5.6
103	Outside diameter, max	Level IV	3.2	Gage
104	Bead cracked	Level IV	3.2	Visual
105	Gold plating	Note 1	3.2	4.5.9
Minor				
201	Length of electrode below surface of header	Level V	3.2	Gage
202	Thickness of header	Level V	3.2	Gage
203	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:
1. Sampling and rejection shall be in accordance with MIL-G-45204.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-D-70826A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Bridge Plug and Ignition Assembly	1 OF 1	12550787
			NEXT HIGHER ASSEMBLY: 12550926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Spot charge missing	Level IV	3.2	Visual
102	Cracks or voids in spot change	Level IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTON

CLASSIFICATION OF CHARACTERISTICS

MIL-D-70826A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Charged cup	1 OF 1	9257046
			NEXT HIGHER ASSEMBLY: 12550926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Height of total charge	Level IV	3.2	Gage
102	Moisture content of lead azide	4.4.3.1	3.3.1	4.5.2.1
103	Moisture content of HMX	4.4.3.1	3.3.2	4.5.2.2
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Detonator Assembly	1 OF 1	12550926
			NEXT HIGHER ASSEMBLY: Final

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Cup cracked or split	100%	3.2	Visual
b	Explosive on exterior of cup	100%	3.2	Visual
Major				
101	Length of electrode	Level IV	3.2	Gage
102	Length, crimped end to base	Level IV	3.2	Gage
103	Depth to spherical radius	Level VI	3.2	Gage
104	Large diameter	Level IV	3.2	Gage
105	Spherical radius missing, deformed or scratched	Note 2	3.2	Visual
106	Anvil spherical radius verification	Level IV	3.2	Gage
107	Sealant not present 360 degrees	Level IV	3.2	Visual
108	Decrease of sealant on electrode beyond sealed area	Level IV	3.2	Visual
109	Waterproofness test	50-0-1 Note 1	3.5	4.5.3.2
110	Functioning output (dent test)	200-1-2 Note1	3.4	4.5.3.1
111	Environmental test			
	a)Resistance between electrode and cup	100%	3.2	4.5.4.1a
	b)Cold temperature conditioning up	25-0-1	3.2	4.5.4.2
	c)Hot temperature conditioning	25-0-1	3.2	4.5.4.3
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES: 1. Any detonator that fails to function on first attempt or fails to produce the required output shall be classed defective.
2. Tooling dimensions are verified prior to manufacturing each lot and at the end of each lot.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.7	Microdetonator Clip Assembly, (After Assembly of Detonator)	SHEET 1 OF 1	9326705
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
special				
a	Detonator not shorted	100%	3.7	Current limited ohmmeter (1 milliamp, max.)
Major				
101	Detonator electrode protruding above flush of clip assembly	100%	3.7	Visual
102	Detonator cup protruding above flush of clip assembly	100%	3.7	Visual
103	Detonator inverted in clip	100%	3.7	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Sealed Bag	1 OF 1	See paragraph 5.1
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	None defined			
Major				
101	Heat seal test	Note 1	3.2	4.5.10
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

1. Sampling and rejection shall be in accordance with MIL-P-116.

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